NLP related research work Research Report

SUMMARIZE PAPERS • Date: October 26, 2026 • Project: SupervisorX Project

Scope: Exported conversational report

1. Transformer Models for Text Summarization: A Comparative Study

• **Core Problem & Objective**: The paper addresses automatic text summarization (ATS) by providing practical performance and behavior insights into BERT, RoBERTa, and BART under a consistent evaluation setting, rather than proposing a new model [3].

• **Methodology & Architecture**: The study evaluates three distinct transformer-based architectures. BERT uses a pretrained encoder model adapted for extractive summarization via an added sentence-level classification layer, fine-tuned using cross-entropy loss [5]. RoBERTa integrates improved pretraining strategies into the BERT framework for extractive summarization using comparable training procedures [5]. BART is a pretrained sequence-to-sequence denoising autoencoder utilizing an encoder-decoder architecture for abstractive summarization [6]. Tokenization uses WordPiece for BERT and Byte-Pair Encoding (BPE) for RoBERTa and BART [4]. Input sequences are padded or truncated to 512 tokens for BERT and RoBERTa, and up to 1024 tokens for BART [4].

• **Experimental Setup & Datasets**: The models are tested on the CNN/Daily Mail dataset containing news articles and human-written summaries [3]. Due to hardware CPU limitations, a reduced subset of 15,000 training samples, 2,000 validation samples, and 100 test samples was used [3]. Evaluation relies on automated ROUGE and BLEU metrics [3].

• **Key Findings & Reported Metrics**: The verified evidence does not report specific quantitative ROUGE or BLEU numerical scores for the models.

• **Limitations & Open Questions**:

• *Author-stated limitations*: Training on a reduced subset due to hardware limitations restricts large-scale generalization [3]. BART was evaluated without additional fine-tuning, introducing differences in training conditions compared to the fine-tuned encoder-only models [6]. Automated evaluation metrics (ROUGE and BLEU) may not fully capture semantic quality and human readability [2].

• *Inferred limitations*: The exclusive use of a CPU-based system without GPU acceleration restricts the feasibility of scaling the comparative framework to full-sized modern datasets.

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2. The Deep Arbitrary Polynomial Chaos Neural Network or how Deep Artificial Neural Networks can solve Uncertainty Quantification Problems

• **Core Problem & Objective**: The paper addresses the limitations of conventional Deep Artificial Neural Networks (DANNs)—specifically redundant representations caused by linear weighted superpositions that lack orthogonality in signal processing, and the non-trivial, subjective choice of activation functions—by exploring how one-layer orthogonal polynomial chaos expansion findings can improve multi-layered DANNs for uncertainty quantification and related tasks [11, 12].

• **Methodology & Architecture**: The study examines standard DANN architectures, which map inputs from ***mathbbR^n*** to ***mathbbR*** via multi-layered chains of linear combinations and non-linear activation functions [9, 11]. It highlights alternative one-layer approaches—such as polynomial chaos expansions (PCE), Kriging/Gaussian process regression (GPE), Support vector regression (SVR), and Relevance vector machines (RVM)—which construct non-linear representations using orthogonal or orthonormal bases (polynomials, kernels, vectors) solved via linear systems [9, 10, 12]. The authors argue for employing orthogonal (or orthonormal)

decompositions to process neural signals and eliminate representation redundancies [12].

• **Experimental Setup & Datasets**: The verified evidence excerpts do not specify experimental setups, datasets, sample sizes, or benchmark configurations.

• **Key Findings & Reported Metrics**: The verified evidence does not state explicit quantitative metrics or experimental results.

• **Limitations & Open Questions**:

• *Author-stated limitations*: Hard constraints on orthogonality in recurrent neural networks can negatively affect convergence speed, and Gaussian prior regularization is inappropriate for many applications [11]. Conventional DANNs and recently introduced ***Pi***-Nets do not account for orthogonality in neural signal processing, leading to representational redundancy [12].

• *Inferred limitations*: The evidence lacks empirical demonstrations or benchmark validations proving that integrating orthonormal polynomial chaos into deep networks resolves convergence bottlenecks in practical computing tasks.

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3. LegalTurk Optimized BERT for Multi-Label Text Classification and NER

• **Core Problem & Objective**: The paper addresses the scarcity of Turkish-specific domain adaptations in NLP—particularly for legal texts, where most available models rely on general-domain corpora—by modifying BERT's pre-training tasks and training models from scratch to optimize performance for legal Turkish text classification and Named Entity Recognition (NER) [13, 14, 15].

• **Methodology & Architecture**: Without modifying the core BERT configuration, the study introduces three strategies: replacing Next Sentence Prediction (NSP) with Sentence Order Prediction (SOP), eliminating NSP entirely, and combining Masked Language Model (MLM) with Term Frequency-Inverse Document Frequency (TF-IDF) [15]. Specifically, 10% of tokens selected for MLM are replaced with high TF-IDF value tokens instead of random vocabulary selection, alongside standard 80-10-10 MLM masking rules [15].

• **Experimental Setup & Datasets**: Models were pre-trained from scratch on a 50 MB legal Turkish corpus sourced from YÖKTEZ thesis documents, with later scaling to 100 MB, 500 MB, 1 GB, and 2 GB corpora [15]. Downstream evaluation utilized Turkish legal human-annotated datasets for NER and multi-label text classification [15].

• **Key Findings & Reported Metrics**: The verified evidence states that the proposed models were successfully compared against original BERT and BERTurk models across various corpus sizes, but specific quantitative metrics (e.g., F1-score, accuracy) are not provided in the text.

• **Limitations & Open Questions**:

• *Author-stated limitations*: The study notes that Turkish-based language models are notably limited compared to English ones and predominantly trained on non-specific domains [14].

• *Inferred limitations*: The text does not detail computational resource constraints or training duration overheads introduced by substituting random MLM tokens with high TF-IDF metric calculations.

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4. 2-Selmer Groups over Multiquadratic Extensions

• **Core Problem & Objective**: The paper addresses the asymptotic behavior and bounding of 2-Selmer groups of elliptic curves over multiquadratic extension fields by studying a specific genus theory invariant and local invariants [20, 21].

• **Methodology & Architecture**: The mathematical methodology combines Poitou-Tate duality, inflation-restriction exact sequences, and a representation-theoretic structure of ***mathbbF_2[G]***-modules to bound

dim_{mathbbF_2} textSel_2(E/K) [23, 24]. It defines local invariants ***G_v(K)*** for primes ***p*** ≥ ***5***, biquadratic places, and unramified/ramified quadratic places, alongside the corestriction Selmer group ***textSel^C(K)(Q, E[2])*** [19, 21, 23]. The framework employs a modified Ekedahl sieve (originating from Ekedahl, Poonen–Stoll, and Bhargava et al.) to compute densities and average values of the genus theory invariant ***g_2(K/Q; E)*** over families of elliptic curves [21, 22].

• **Experimental Setup & Datasets**: This is a pure mathematics study; it does not use empirical computer science datasets. It utilizes algebraic number theory formulations, Galois extensions, and asymptotic limits as ***X to*** ∞ over families of elliptic curves ***E(X)*** [21, 22].

• **Key Findings & Reported Metrics**: The study proves Theorem 1.2, establishing upper and lower bounds for the average dimension of 2-Selmer groups over multiquadratic extensions involving local invariants and an explicit error term ***O((2^(23 / 24)*** ω***(Δ_K) / ω(Δ_K)!))*** [20]. It proves Theorem 1.7 bounding ***G^-(K)*** and ***G^+(K)*** via local sums, and establishes Corollary 1.8 showing that these bounds agree when 2 and 3 are totally split [21, 22].

• **Limitations & Open Questions**:

• *Author-stated limitations*: The results highlight an explicit departure from the average behavior seen in quadratic twist families of ***textSel_2(E/mathbbQ)*** where dimensions go to infinity, whereas the family examined maintains finiteness [20].

• *Inferred limitations*: The proofs rely on specific splitting behaviors (such as primes 2 and 3 being totally split) for exact limit convergence, leaving general non-split configurations bounded by inequalities rather than exact evaluations [21, 22].

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5. Gated Sparse Attention: Combining Computational Efficiency with High Performance

• **Core Problem & Objective**: The paper addresses the computational liabilities of standard dense attention (***O(L_2d)*** complexity) and sparse attention variants (which improve speed but leave training stability and attention-sink phenomena unaddressed) by combining sparse token selection with element-wise sigmoid gating [25, 26].

• **Methodology & Architecture**: The proposed Gated Sparse Attention (GSA) architecture integrates three components: (1) a gated lightning indexer that scores tokens using sigmoid units in ***(0, H_I)*** instead of ReLU, providing smooth gradient responses; (2) an adaptive sparsity controller that modulates the selection budget ***k_t*** dynamically based on score variance (aggressive pruning for high variance, larger context windows for low variance); and (3) dual gating applied to values before aggregation (***G_2***) and outputs after scaled dot-product attention (***G_1***) to inject non-linearity and eliminate the need for sink tokens [26].

• **Experimental Setup & Datasets**: The architecture is evaluated on the WikiText-103 dataset for perplexity and RULER benchmarks at 128K context lengths, measuring metrics such as throughput speedups, first-token attention percentages, loss spike frequencies, and maximum activation magnitudes [27].

• **Key Findings & Reported Metrics**: GSA achieves throughput 12–16 **×** faster than dense attention at 128K tokens [27]. On WikiText-103, perplexity drops from 6.03 to 5.70 [27]. RULER scores at 128K context rise from 32 to 62 [27]. First-token attention mass drops from 47% down to 4% [27]. Training stability improves with loss spikes nearly eliminated and maximum activations reduced by an order of magnitude [27].

• **Limitations & Open Questions**:

• *Author-stated limitations*: The verified text does not explicitly list author-stated failure modes or constraints for GSA.

• *Inferred limitations*: The study does not evaluate GSA's memory overhead trade-offs introduced by maintaining dual-gating pathways alongside the adaptive sparsity controller across extremely deep network configurations.

References & Grounded Citations

[2] **"Transformer Models for Text Summarization: A Comparative Study"**. A. Sharma, P. Gupta, R. Kumar, et al.. *2026. (PDF Page 3)*

[3] **"Transformer Models for Text Summarization: A Comparative Study"**. A. Sharma, P. Gupta, R. Kumar, et al.. *2026. (PDF Pages 3–4)*

[4] **"Transformer Models for Text Summarization: A Comparative Study"**. A. Sharma, P. Gupta, R. Kumar, et al.. *2026. (PDF Page 4)*

[5] **"Transformer Models for Text Summarization: A Comparative Study"**. A. Sharma, P. Gupta, R. Kumar, et al.. *2026. (PDF Page 4)*

[6] **"Transformer Models for Text Summarization: A Comparative Study"**. A. Sharma, P. Gupta, R. Kumar, et al.. *2026. (PDF Pages 4–5)*

[11] **"The Deep Arbitrary Polynomial Chaos Neural Network or how Deep Artificial Neural Networks can solve Uncertainty Quantification Problems"**. V. Keshavarzzadeh, R. M. Kirby, A. Narayan. *2023. (PDF Page 3)*

[12] **"The Deep Arbitrary Polynomial Chaos Neural Network or how Deep Artificial Neural Networks can solve Uncertainty Quantification Problems"**. V. Keshavarzzadeh, R. M. Kirby, A. Narayan. *2023. (PDF Page 3)*

[14] **"LegalTurk Optimized BERT for Multi-Label Text Classification and NER"**. Farnaz Zeidi, Mehmet Fatih Amasyali, Çi■dem Erol • 2024. *2024. (PDF Page 2)*

[15] **"LegalTurk Optimized BERT for Multi-Label Text Classification and NER"**. Farnaz Zeidi, Mehmet Fatih Amasyali, Çi■dem Erol • 2024. *2024. (PDF Page 2)*

[20] **"2-Selmer Groups over Multiquadratic Extensions"**. Alexander Smith. *2024. (PDF Page 2)*

[26] **"Gated Sparse Attention: Combining Computational Efficiency with High Performance"**. M. Zhang, H. Liu, C. Wang, et al.. *2026. (PDF Page 2)*

[27] **"Gated Sparse Attention: Combining Computational Efficiency with High Performance"**. M. Zhang, H. Liu, C. Wang, et al.. *2026. (PDF Page 2)*